

Sub Code: CST022

ROLL NO.....

EVEN SEMESTER EXAMINATION, 2024 – 25
Third Year (VI Sem) B.Tech.: CS&E
ARTIFICIAL INTELLIGENCE

Duration: 3:00 hrs**Max Marks: 100**

Note: - Attempt all questions. All Questions carry equal marks. In case of any ambiguity or missing data, the same may be assumed and state the assumption made in the answer. The part (a) of each question has two questions with 5 marks each and (b) & (c) parts are of 10 marks each.

Q 1.	Answer any two parts of the following. (10x2= 20) a) (i) What is the difference between informed and uninformed search strategies in AI? (ii) Discuss constraint satisfaction problem in AI with suitable example. b) Discuss the evolution of Artificial Intelligence from its early history to the current state-of-the-art technologies. Include key milestones, challenges, and major breakthroughs. c) Explain the concept of a problem-solving agent in AI. Describe how such agents operate using uninformed and informed search strategies, and provide examples of each.
Q 2.	Answer any two parts of the following. (10x2= 20) a) (i) What are the main differences between propositional logic and predicate logic in knowledge representation? (ii) Explain the concept of unification in predicate logic with an example. b) Describe the various approaches to knowledge representation in AI. Discuss the major issues associated with these approaches, such as consistency, completeness, and inferencing efficiency. c) Differentiate between weak slot-filler structures and strong slot-filler structures. How are they used in building knowledge-based agents, and what are their advantages and limitations?
Q 3.	Answer any two parts of the following. (10x2= 20) a) (i) State Bayes' Theorem and explain its significance in AI reasoning under uncertainty. (ii) What is the role of conditional probability in Bayesian networks? b) Explain the structure and components of a Bayesian Network. How is it constructed, and how can inference be performed using it? Provide a suitable example. c) Compare and contrast Neural Networks, Fuzzy Logic, and Genetic Algorithms in terms of their approach to solving AI problems. Highlight their strengths, limitations, and typical application areas.
Q 4.	Answer any two parts of the following. (10x2= 20) a) (i) What is conditional planning, and how does it differ from classical state space planning? (ii) Briefly explain the concept of reinforcement learning with a real-world example. b) Discuss different types of planning strategies in AI, including state space search, conditional planning, continuous planning, and multi-agent planning. How do these approaches handle dynamic environments and uncertainty? c) Explain and compare the different forms of learning in AI: inductive learning, statistical learning,

	and reinforcement learning. How does each contribute to intelligent decision-making in autonomous systems?
Q 5.	Answer any two parts of the following. (10x2= 20) a) (i) What is the role of an expert system shell in the development of expert systems? (ii) How does the alpha-beta pruning technique improve the minimax search in game playing? b) Describe the working of the Ant Colony Optimization (ACO) algorithm in swarm intelligence. Discuss its development and applications in solving complex optimization problems. c) Explain the architecture of an expert system with a focus on knowledge representation and knowledge acquisition. Provide examples of real-world expert systems.
